HDOC Day-49

Aim: Write a C program to find the pivot integer between 1 and n using a function and pointer.

Program:

```
#include <stdio.h>

void pivot_integer(int *n)
{
    int x, left, right, i;

    for (x = 1; x <= *n; x++)
    {
        left = 0;
        right = 0;

        for (i = 1; i <= x; i++)
            left += i;

        for (i = x; i <= *n; i++)
            right += i;

        if (left == right)
        {
            printf("Pivot integer: %d\n", x);
            return;
        }
    }
    printf("-1\n");
}

int main()
{
    int n;
    printf("Enter a positive integer: ");
    scanf("%d", &n);

    if (n <= 0)
        printf("Please enter a positive integer.\n");
    else
        pivot_integer(&n);

    return 0;
}
```

Sample Outputs:

Input: n = 8
Output: Pivot integer: 6

Input: n = 1
Output: Pivot integer: 1

Input: n = 4
Output: -1